Lever Test · Knowledge Organiser

Learning goal: I can test how the distance from the pivot changes the effort needed to lift a load.

Today I Can

I can keep the load and pivot position the same while changing the effort distance.

I can record the effort needed (in coins/masses or newtons) at three distances.

I can use my results to state how effort distance affects the force needed.

Must-Know Words

effort distance: the distance from the pivot to where the effort push is applied

load: the weight or mass being lifted by the lever

independent variable: the one thing we change on purpose (the effort distance from the pivot)

pattern: how one measurement changes when we change another

Core Science Steps

First, check the controls: the pivot stays fixed and the same load sits 5 cm from the pivot every time.

Second, compare the three effort distances: at 5 cm it took 8 coins, at 10 cm it took 4 coins, and at 20 cm it took 2 coins.

Third, state the pattern: as the distance from the

pivot to the effort increases, the effort needed decreases.

Keep the Science Clear

Quote at least two distances and their coin counts when you write your conclusion.

Our ruler has mass of its own, so if the pivot is not near the middle, the ruler's own weight also pulls on one side.